Immanuel Tampubolon



AI Engineer Intern · LLM agents, RAG and multi-agent systems

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Seeking an internship. Available onsite, including relocation abroad, or remote.

PROFILE

Computer Technology student at Institut Teknologi Del. Most of my work this past year has gone into LLM systems that have to hold up on real input rather than on a demo prompt: a coding agent that runs a project's own test suite to check its own patches, a three-agent anomaly-detection pipeline that falls back to rules when the LLM API is throttled, and an Indonesian-language RAG security assistant that I evaluated with RAGAS and am now extending for my thesis. The parts I find hardest, and most interesting, are retrieval quality and knowing whether the system actually works. I am looking for an AI Engineer internship where I can build LLM features alongside a team.

TECHNICAL SKILLS

LLM and agents	RAG pipelines, multi-agent orchestration, tool and function calling, self-correction loops, prompt engineering, LangChain
Models and APIs	Groq (Llama 3, 3.1, 3.3), Hugging Face Inference Providers, Qwen, Gemini API, Cohere
Retrieval and ML	Embeddings (Cohere, sentence-transformers), Supabase pgvector, cosine-similarity search, scikit-learn (Isolation Forest, Random Forest)
Evaluation	RAGAS, Langfuse tracing, Server-Sent Events streaming
Languages	Python, TypeScript, JavaScript, Go, PHP. Foundational Java and C
Backend and frontend	FastAPI, Flask, Node.js, Laravel, Go. React, Next.js, React Native, Tailwind CSS
Data and DevOps	PostgreSQL, MySQL, MongoDB, Redis Streams, Docker, Nginx, Git, Vercel, Railway, Render

AI ENGINEERING PROJECTS

Sentinel. Autonomous self-correcting coding agent

Personal project, 2026

Python, FastAPI, Llama 3.1 via Hugging Face Inference Providers, sentence-transformers, Docker, Langfuse, Next.js

- An agent that closes the bug-fix loop on its own. The model proposes a patch, the project's real test suite runs in an isolated sandbox (Docker, or a virtualenv provisioned per project), and failing test output goes back to the model for up to three further attempts.
- Every fix is grounded in repository context through a retriever built on sentence-transformers embeddings and cosine similarity, so the model reasons over related modules instead of the target file alone.
- Takes any external repository by GitHub clone or ZIP upload and installs its real dependencies automatically. I validated it end to end against the open-source Flask codebase.
- Run-level tracing goes to Langfuse, and a Next.js interface streams the agent's reasoning over Server-Sent Events as it retrieves, generates and runs tests.

AnomalyGuard. Multi-agent network anomaly detection

Personal project, 2026

Python, Flask, Redis Streams, Groq LLM with function calling, scikit-learn, Next.js, Recharts

- Three agents (detection, decision, response) communicate over Redis Streams. An Isolation Forest and Random Forest ensemble scores traffic, and an LLM agent investigates the high-risk cases using function-calling tools for incident history, active-incident count and traffic baseline.
- Only high-risk events reach the LLM, and a rate-limit circuit breaker drops back to rule-based decisions when the API is throttled or unavailable. Keeping the system useful under quota limits turned out to be most of the work.
- A Next.js and Recharts dashboard pulls live data from the three backend services across eight views, including LLM quota health and the dead-letter queue.

CyberGuard AI. RAG security assistant for Indonesian users

Team of three, basis of thesis

FastAPI, LangChain, Cohere embeddings, Qwen, Supabase pgvector, RAGAS, Langfuse, Next.js, Railway

- I built the retrieval pipeline end to end: ingestion, Cohere embeddings, pgvector storage, and LangChain retrieval and generation over 1,292 documents drawn from MITRE ATT&CK and a custom set of 50 or more Indonesian scam case studies.
- Added multi-turn conversational memory and reworked retrieval where sources overlapped. Answer quality measured with RAGAS, requests traced with Langfuse.

- Also shipped a malicious-URL verification tool and an interactive security learning module, with the FastAPI backend deployed on Railway.
- For my thesis I am splitting it into separate retriever, verifier and responder agents, with LoRA fine-tuning for Indonesian-language digital-fraud detection.

DevPulse. AI code review SaaS *Personal project*

Go, Next.js, Groq (Llama 3.3), GitHub OAuth, Vercel, Railway

- Reviews GitHub pull requests automatically with Llama 3.3, flagging exposed secrets and other security leaks, and writes its reports in both English and Indonesian.
- Go backend with GitHub OAuth sign-in and a Next.js frontend, running in production on Vercel and Railway.

Apotek Century. Pharmacy mobile app with AI assistant *Full-stack project*

React Native (Expo), Laravel 11, FastAPI, Groq (Llama 3), Midtrans

- An AI medical assistant running on Llama 3 checks drug interactions inside a working pharmacy app, alongside real-time chat with a pharmacist.
- I also built the commerce side: catalogue with category filters and discounts, cart and Midtrans checkout, prescription upload, and an admin dashboard.

OTHER PROJECTS

Connect+, a five-service microservices campus platform deployed across three machines with a MongoDB replica set, Docker, Nginx and HAProxy. **Neo-Vanguard**, an AI-literacy and cybersecurity education platform in React and TypeScript. **Tanah Batak**, a cultural-heritage site in React, TypeScript and Go. **SiPerpus**, a library management system in Laravel. **UNTHOPEDINE**, a coffee-shop site with WhatsApp ordering.

EDUCATION AND ORGANISATION

Institut Teknologi Del, Diploma (D3, A.Md.) in Computer Technology *2024 to present, Toba*

Focus on software development, computer networking and embedded systems. Thesis in progress: developing CyberGuard AI into a modular multi-agent system with language-model fine-tuning for Indonesian-language digital fraud detection.

Himpunan Mahasiswa Teknologi Komputer, Facilities and Infrastructure Division *2025 to present*

CERTIFICATIONS, TRAINING AND COMPETITIONS

AI AND MACHINE LEARNING

AI Engineer for Millennials, Komdigi DTS	*Feb 2026*
AI Security Fundamentals, Komdigi and Microsoft DTS	*Feb 2026*
Intro to Machine Learning, Kaggle	*Apr 2026*
GenAI-Powered Data Analytics Job Simulation, Tata and Forage	*Sep 2026*

NETWORKING AND SECURITY

Introduction to Cybersecurity, Cisco Networking Academy	*Feb 2026*
HCIA-Datacom V1.0 Course, Huawei ICT Academy	*Nov 2025*
HCIA-Storage V5.0 Course, Huawei ICT Academy	*Oct 2025*

SOFTWARE ENGINEERING

Software Engineer Role Certification, HackerRank	*Sep 2026*
Software Engineering Job Simulation, Commonwealth Bank and Forage	*Apr 2026*
Memulai Pemrograman dengan Java, Dicoding	*Feb 2026*
Memulai Pemrograman dengan C, Dicoding	*Feb 2026*

COMPETITIONS AND BOOTCAMPS

QuantifAI Bootcamp, AI and 5D BIM Master Challenge, Glodon Indonesia with the Ministry of Public Works	*Sep 2026*
Web Design Competition, participant, HIMA-RPL Politeknik Negeri Indramayu	*Apr 2026*

VIRTUAL INTERNSHIP PROGRAMMES

Artificial Intelligence Intern, CodeAlpha *Oct 2026*

One-month remote programme building practical AI assignments and projects.

Python Programming Intern, Oasis Infobyte *Oct 2026*

A month of hands-on Python project work.

C/C++ Development Intern, Codveda Technologies *Feb 2026*

Remote programme covering C and C++ development, testing and optimisation.

Languages. Indonesian (native), English.